Slippage Desk

The options agent that measures its own execution.

Alpaca AI Trading Agents Hackathon · paper account
PA343VC6LL3T

The hard part is not choosing *what* to trade.

It is getting the price the edge assumed. That gap is invisible unless you score every fill against the mid it was priced at.

A win banks	+$31.50
The stop costs	−$78.74
Breakeven win rate	71.4%
Delta-implied OTM rate	76%

The entire edge is 4.6 percentage points, worth $5.04 a contract. Crossing the bid/ask cost $1.38 of that, 27% of the edge. Every figure derived from config and from the 43 contracts actually filled.

"LLM proposes, deterministic gates authorise."

That architecture is table stakes for an agent allowed near a live account. This project has it, and it is not the claim.

The claim is narrower and harder to fake: this desk knows, per fill, what its execution actually cost, and it trades differently because of it.

The claim is that execution quality is measurable in four days, and strategy edge is not.

Score every fill against the mid it was priced at.

- → Capture ratio kept per `(underlying, tenor, delta band, time of day)`
- → Shrunk toward 1.0, so one lucky fill cannot make a bucket look expert
- → Buckets that surrender too much credit are **refused**
- → The same memory sets how far to cross on the next order

Buckets that rarely fill lean toward crossing. Buckets that fill readily hold out for mid.
The agent closes a loop on its own execution rather than just logging it.

The model has strictly negative authority.

It may shrink a trade or veto it. Nothing else.

- → Fifteen deterministic gates run first; sizing is a fixed formula
- → The clamp **multiplies**, never assigns. A model returning 5.0 is treated as 1.0
- → It cannot pick a strike, change an expiry, or widen risk
- → Timeout, refusal, malformed JSON, NaN: every failure path is a veto

The failure path is the design. Ask any LLM-driven system what it does when the model is unreachable. This one refuses: failure degrades to trading less, never to trading unsupervised.

Early assignment.

A short call goes in-the-money. An ex-dividend falls before expiry. Capturing the dividend beats the remaining extrinsic, so the holder exercises early.

You are assigned 100 shares per contract and carrying an overnight gap you never agreed to. The spread's defined risk was defined against expiry, not against being assigned three days before it.

So assignment is a gate, checked against the ex-dividend calendar before every entry, not a footnote.

Three surfaces, deliberately separate jobs.

SURFACE	JOB
`CLI`	Execution and book verification. Nine endpoints.
`Python SDK`	Chains with greeks and IV in one call.
`MCP server`	Read-only research for the advisor.

Execution never routes through MCP, and that is enforced rather than intended: the server launches without the `trading` toolset, so no order tool exists to call. A stdio subprocess must never sit between the agent and its stops.

`15 / 15 surfaces responding, live on the dashboard.`

Not a screenshot. A reproducible artefact.

CONSIDERED	SPREADS FILLED	CREDIT CAPTURED	LOST TO EXECUTION
2,368	36	98.3%	$60
candidates evaluated	broker-paired, all fifteen gates	broker-verified	what execution actually cost

`data/proof.json` carries every fill, the mid it was priced against, the credit received, per-bucket capture, every bucket refused, and a sha256 over the payload.

Sourced from Alpaca's own activity log. Anyone holding the account ID can reproduce it.

What the account actually did.

P&L over the competition window	−$332.62
As a share of starting capital	−0.33%
Against a daily loss limit of	2.00%
Theoretical credit captured	98.3%

Every position was defined risk, so the maximum loss was known before each order was sent. The stops closed the losers near 2x credit rather than at full width, the daily loss limit was never reached, and the portfolio risk ceiling never bound.

A short-premium book in a rising tape is the market this strategy is built to lose in. It lost under half a percent doing it.

What we claim, and what we don't.

We do not claim the strategy is profitable. Four sessions cannot establish that, and treating this window as one would be reading noise, not signal.

We claim the agent measured its own execution quality and acted on it, and we ship the evidence.

> *"Spot sits only 0.8% below the 721 short strike after a 1.2% down day, and the 0.75 credit on a 5-wide spread does not pay for a 2-DTE gamma gap."*
>
> The advisor, declining a trade the gates had approved.

Slippage Desk

Choosing what to trade is the easy half.
This one learns whether it can get the price.

```
github.com/priyanshshahh/slippage-desk
slippage-desk.vercel.app
Alpaca paper account PA343VC6LL3T
```

Next: cross-venue routing on the same memory, and per-bucket sizing rather than per-bucket veto.

Built with Claude Code.